

Chapter 6: Linear Transformations

Chapter overview

A linear transformation is a function between vector spaces that preserves the operations defining linear structure. This chapter develops four complementary descriptions:

$$\text{formula} \longleftrightarrow \text{matrix} \longleftrightarrow \text{kernel and range} \longleftrightarrow \text{geometry}.$$

Chapter objectives. By the end of the chapter, students should be able to test linearity, find images and preimages, calculate kernels and ranges, use rank-nullity, construct transformation matrices, change bases, recognize similar matrices, and apply matrices to geometric transformations.

Original contexts

The examples use original numerical values and selectively draw on travel, hiking, badminton, driving, music, and Spurs-inspired themes. Fictional routes, scores, and coordinates are teaching models.

1 Introduction to Linear Transformations

Domain, codomain, image, preimage, and range

For $T : V \rightarrow W$, V is the domain and W is the codomain. If $T(\mathbf{v}) = \mathbf{w}$, then $\mathbf{w}$ is the image of $\mathbf{v}$ and $\mathbf{v}$ is a preimage of $\mathbf{w}$. The range is the set of all attainable images.

Travel-profile transformation

Define

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(x, y) = (2x - y, x + 3y).$$

For $(3, 1)$,

$$T(3, 1) = (6 - 1, 3 + 3) = (5, 6).$$

To find the preimage of $(5, 6)$, solve

$$2x - y = 5, \quad x + 3y = 6.$$

The first equation gives $y = 2x - 5$. Substitution gives

$$x + 3(2x - 5) = 6 \implies 7x = 21 \implies x = 3,$$

and $y = 1$.

Linearity test

A map $T : V \rightarrow W$ is linear when

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{u}) = cT(\mathbf{u})$$

for every $\mathbf{u}, \mathbf{v} \in V$ and scalar c .

For $T(x, y) = (2x - y, x + 3y)$,

$$\begin{aligned}T(\mathbf{u} + \mathbf{v}) &= T(u_1 + v_1, u_2 + v_2) \\&= (2u_1 - u_2, u_1 + 3u_2) + (2v_1 - v_2, v_1 + 3v_2) \\&= T(\mathbf{u}) + T(\mathbf{v}),\end{aligned}$$

and

$$T(c\mathbf{u}) = (2cu_1 - cu_2, cu_1 + 3cu_2) = cT(\mathbf{u}).$$

Thus T is linear.

Fast test

Every linear transformation satisfies $T(\mathbf{0}) = \mathbf{0}$. Therefore, $F(x, y) = (x + 1, y)$ is not linear because $F(0, 0) = (1, 0)$.

A basis determines the entire transformation

Suppose

$$T(\mathbf{e}_1) = (9, 1), \quad T(\mathbf{e}_2) = (20, -3), \quad T(\mathbf{e}_3) = (21, 4).$$

Since

$$\begin{aligned}(2, -1, 3) &= 2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3, \\ T(2, -1, 3) &= 2T(\mathbf{e}_1) - T(\mathbf{e}_2) + 3T(\mathbf{e}_3) \\ &= 2(9, 1) - (20, -3) + 3(21, 4) \\ &= (61, 17).\end{aligned}$$

Every $m \times n$ matrix A defines a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$. For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix},$$
$$A\mathbf{v} = \begin{bmatrix} 0 \\ 7 \\ -4 \end{bmatrix}.$$

Differentiation $D : P_3 \rightarrow P_2$ and the definite-integral functional $L(p) = \int_0^1 p(x) dx$ are linear because derivative and integral rules preserve sums and scalar multiples.

Domain	Input vector space.
Codomain	Declared target vector space.
Image	Output $T(\mathbf{v})$.
Preimage	Input mapped to a chosen output.
Range	Set of all images.
Linear transformation	Map preserving addition and scalar multiplication.

2 Kernel and Range

Two fundamental subspaces

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}, \quad \text{range}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}.$$

The kernel is a subspace of the domain, and the range is a subspace of the codomain.

Projection onto the xy -plane

For

$$T(x, y, z) = (x, y, 0),$$

$T(x, y, z) = \mathbf{0}$ exactly when $x = y = 0$, so

$$\ker(T) = \{(0, 0, z) : z \in \mathbb{R}\} = \text{span}\{(0, 0, 1)\}.$$

The outputs fill the xy -plane:

$$\text{range}(T) = \text{span}\{(1, 0, 0), (0, 1, 0)\}.$$

Detailed rank-nullity calculation

Let

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

Then

$$\begin{aligned} A &\xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -2 & 0 \end{bmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -2 \end{bmatrix}. \end{aligned}$$

There are three pivots, so $\text{rank}(A) = 3$.

Solving $A\mathbf{x} = \mathbf{0}$ gives

$$x_1 + 2x_3 + x_4 = 0, \quad x_2 - x_3 + 2x_4 = 0, \quad -x_3 - 2x_4 = 0.$$

Let $x_4 = t$. Then

$$x_3 = -2t, \quad x_2 = -4t, \quad x_1 = 3t,$$

so

$$\ker(T) = \text{span}\{(3, -4, -2, 1)\}, \quad \text{nullity}(T) = 1.$$

Rank-nullity theorem

If $T : V \rightarrow W$ and V is finite dimensional, then

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

Here,

$$3 + 1 = 4 = \dim(\mathbb{R}^4).$$

The map is onto $\mathbb{R}^3$ because its rank is 3, but it is not one-to-one because its nullity is positive.

A linear map is one-to-one iff $\ker(T) = \{\mathbf{0}\}$. It is onto iff $\text{rank}(T) = \dim(W)$. A one-to-one and onto linear map is an isomorphism. For example,

$$C : P_2 \rightarrow \mathbb{R}^3, \quad C(a + bx + cx^2) = (a, b, c)$$

is an isomorphism with inverse $(a, b, c) \mapsto a + bx + cx^2$.

3 Matrices for Linear Transformations

Standard matrix theorem

For $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)]$$

and $T(\mathbf{x}) = A\mathbf{x}$.

Let

$$T(x, y, z) = (x + 2y - z, 3x - y + 4z).$$

The standard basis images are

$$T(\mathbf{e}_1) = (1, 3), \quad T(\mathbf{e}_2) = (2, -1), \quad T(\mathbf{e}_3) = (-1, 4),$$

so

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix}.$$

For $(2, -1, 3)$,

$$A \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \end{bmatrix}.$$

Composition and inverse

If T_1 has matrix A_1 and T_2 has matrix A_2 , then $T_2 \circ T_1$ has matrix A_2A_1 . With

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$
$$A_2A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \quad A_1A_2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}.$$

The products differ because applying transformations in different orders usually gives different results. For

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

$\det(A) = 1$, so

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

and

$$T^{-1}(u, v) = (u - v, -u + 2v).$$

Nonstandard bases

Let $T(x, y) = (x + y, 2x - y)$, with

$$B = \{(1, 1), (1, -1)\}, \quad C = \{(1, 0), (1, 1)\}.$$

Because

$$T(1, 1) = (2, 1) = 1(1, 0) + 1(1, 1),$$

and

$$T(1, -1) = (0, 3) = -3(1, 0) + 3(1, 1),$$

$$[T]_{C \leftarrow B} = \begin{bmatrix} 1 & -3 \\ 1 & 3 \end{bmatrix}.$$

Relative to $\{1, x, x^2, x^3\}$ and $\{1, x, x^2\}$, differentiation has matrix

$$[D] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

4 Transition Matrices and Similarity

Transition matrix

If B and C are basis matrices, then

$$[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B, \quad P_{C \leftarrow B} = C^{-1}B.$$

For

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

$$C^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix},$$

so

$$P_{C \leftarrow B} = C^{-1}B = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}.$$

Thus

$$\begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Similar matrices

Matrices A and A_B are similar if

$$A_B = P^{-1}AP$$

for an invertible matrix P . They represent the same linear transformation in different bases.

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Then

$$P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Compute

$$AP = \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix},$$

and then

$$P^{-1}AP = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Similar matrices have the same trace, determinant, characteristic polynomial, eigenvalues, and rank. Here both matrices have trace 4 and determinant 3.

The diagonal representation makes powers easy:

$$A^n = P \begin{bmatrix} 3^n & 0 \\ 0 & 1 \end{bmatrix} P^{-1} = \frac{1}{2} \begin{bmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{bmatrix}.$$

Thus

$$A^5 = \begin{bmatrix} 122 & 121 \\ 121 & 122 \end{bmatrix}.$$

5 Applications of Linear Transformations

Common planar matrices

Transformation	Matrix	Coordinate rule
Reflection in y -axis	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	$(x, y) \mapsto (-x, y)$
Reflection in $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$(x, y) \mapsto (y, x)$
Horizontal shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	$(x, y) \mapsto (x + ky, y)$
Scaling	$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$	$(x, y) \mapsto (ax, by)$
Rotation	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$	counterclockwise by θ

Badminton-court shear

The horizontal shear

$$H(1/2) = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

sends (x, y) to $(x + y/2, y)$. Therefore, the rectangle

$$(0, 0), (6, 0), (6, 3), (0, 3)$$

becomes

$$(0, 0), (6, 0), (15/2, 3), (3/2, 3).$$

The upper edge shifts right while the lower edge remains fixed.

Rotation and composition

A 30° rotation sends

$$\begin{bmatrix} 2 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}.$$

If a horizontal stretch by 2 occurs before a 90° rotation, then

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}.$$

The reverse order gives a different matrix.

Area scaling

A planar linear transformation with matrix A multiplies area by $|\det(A)|$. Shears and rotations have determinant magnitude 1, so they preserve area.

A rotation about the z -axis is

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

At 90° ,

$$R_z(90^\circ)(1, 2, 3) = (-2, 1, 3).$$

Chapter terminology

Linear transformation	Preserves addition and scalar multiplication.	Kernel	Inputs mapped to zero.
Range	Attainable outputs.	Rank	Dimension of the range.
Nullity	Dimension of the kernel.	Isomorphism	One-to-one and onto linear map.
Standard matrix	Columns are images of standard basis vectors.	Composition	Successive application of maps.
Transition matrix	Converts coordinates between bases.	Similarity	Relation $A_B = P^{-1}AP$.
Reflection	Mirror transformation.	Shear	Sliding transformation.
Rotation	Rigid turn about an origin or axis.	Area scale factor	$ \det(A) $.